PHYS 331: Stellar Structure and Evolution.

Numerical Modeling of Polytropic Stellar Structures Project.

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Abstract

This project investigates the internal structure of stars by numerically solving the Lane-Emden equation for polytropic stellar models with indices $n = 1.5$, $n = 3.0$, and $n = 4.0$. Using analytical derivations and a Python implementation calibrated to solar mass and radius values, we analyzed how the polytropic index affects physical variables such as density, pressure, and temperature profiles. To evaluate the physical accuracy of these simplified models, we compared the numerical results against realistic stellar simulations from MESA (Modules for Experiments in Stellar Astrophysics) and observational populations from the Gaia space observatory. The results demonstrate that while $n = 1.5$ best represents extended, low-mass convective stars and $n = 4.0$ yields an unphysically condensed structure, the $n = 3.0$ polytrope matches the MESA solar model with exceptional accuracy ($MSE \approx 0.000744$). This confirms the $n = 3.0$ model as a highly effective analytical representation of stable, radiative stars like our Sun.

1 Introduction

The internal structure of stars is determined by the interaction between gravitational forces and pressure support. The equilibrium of these forces, called hydrostatic equilibrium, determines the distribution of physical quantities in a star [1].

Although the solution of complex systems involving nuclear processes and energy transport is a feature of realistic stellar models, simplified models can give useful physical insight. One approach is the polytropic model, where pressure and density are related by a power law [1].

Polytropic models are especially useful for describing different regimes of stars. For example, the case of $n = 1.5$ is a good representation of fully convective stars, while $n = 3$ corresponds to degenerate objects such as white dwarfs. Larger indices correspond to more centrally condensed configurations.

In this work, we solve the Lane–Emden equation numerically and analyze the resulting stellar structures.

2 Theoretical Framework

2.1 Polytropic Relation

The polytropic equation of state is written as

$$P = K\rho^{\gamma} \tag{1}$$

where

$$\gamma = 1 + \frac{1}{n} \tag{2}$$

which gives

$$n = \frac{1}{\gamma - 1} \tag{3}$$

2.2 Derivation of the Lane–Emden Equation

We begin with the equation of hydrostatic equilibrium [1],

$$\frac{dP}{dr} = -\frac{Gm(r)\rho}{r^2} \tag{4}$$

and the equation of mass conservation [1],

$$\frac{dm}{dr} = 4\pi r^2 \rho \tag{5}$$

Using the substitutions for non-dimensionalization,

$$\rho = \rho_c \theta^n \tag{6}$$

and

$$r = \alpha \xi \tag{7}$$

The pressure becomes

$$P = K\rho^{1+\frac{1}{n}} \tag{8}$$

Substituting the density relation,

$$P = K(\rho_c \theta^n)^{1+\frac{1}{n}} \tag{9}$$

which simplifies to

$$P = K\rho_c^{1+\frac{1}{n}} \theta^{n+1} \tag{10}$$

Differentiating with respect to θ,

$$\frac{dP}{d\theta} = K\rho_c^{1+\frac{1}{n}} (n+1)\theta^n \tag{11}$$

Using the chain rule,

$$\frac{dP}{dr} = \frac{dP}{d\theta}\frac{d\theta}{dr} \tag{12}$$

therefore,

$$\frac{dP}{dr} = K\rho_c^{1+\frac{1}{n}} (n+1)\theta^n \frac{d\theta}{dr} \tag{13}$$

Substituting into the hydrostatic equilibrium equation gives

$$K\rho_c^{1+\frac{1}{n}} (n+1)\theta^n \frac{d\theta}{dr} = -\frac{Gm(r)\rho}{r^2} \tag{14}$$

Substituting

$$\rho = \rho_c \theta^n \tag{15}$$

we obtain

$$K\rho_c^{1+\frac{1}{n}}(n+1)\theta^n \frac{d\theta}{dr} = -\frac{Gm(r)\rho_c\theta^n}{r^2} \tag{16}$$

Canceling θ^n,

$$K\rho_c^{1+\frac{1}{n}}(n+1)\frac{d\theta}{dr} = -\frac{Gm(r)\rho_c}{r^2} \tag{17}$$

which gives

$$\frac{d\theta}{dr} = -\frac{Gm(r)}{K\rho_c^{1/n}(n+1)r^2} \tag{18}$$

Multiplying both sides by r^2,

$$r^2\frac{d\theta}{dr} = -\frac{G}{K\rho_c^{1/n}(n+1)}m(r) \tag{19}$$

Define the constant

$$A = \frac{G}{K\rho_c^{1/n}(n+1)} \tag{20}$$

thus,

$$r^2\frac{d\theta}{dr} = -Am(r) \tag{21}$$

Differentiating both sides with respect to r,

$$\frac{d}{dr}\left(r^2\frac{d\theta}{dr}\right) = -A\frac{dm}{dr} \tag{22}$$

Using equation (5), therefore,

$$\frac{d}{dr}\left(r^2\frac{d\theta}{dr}\right) = -A(4\pi r^2\rho_c\theta^n) \tag{23}$$

Now define

$$\alpha^2 = \frac{K(n+1)}{4\pi G}\rho_c^{\frac{1}{n}-1} \tag{24}$$

with

$$r = \alpha\xi$$

Substituting these dimensionless variables,

$$\frac{d}{dr}\left(r^2\frac{d\theta}{dr}\right) = \frac{1}{\alpha^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right) \tag{25}$$

Finally, the Lane–Emden equation is obtained:

$$\boxed{\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right) = -\theta^n} \tag{26}$$

with boundary conditions

$$\theta(0) = 1, \qquad \frac{d\theta}{d\xi}(0) = 0 \tag{27}$$

Thus, the Lane–Emden equation governing the structure of a polytropic sphere is obtained. Its solutions determine important physical properties, such as the density distribution, pressure profile, and radius of the star, for self-gravitating stellar models with different polytropic indices n.

3 Numerical Results

In this section, the numerical solutions of the Lane–Emden equation are presented for different values of the polytropic index, namely $n = 1.5$, $n = 3$, and $n = 4$. The solutions were obtained using a Python program, which is included in the appendix. The numerical method was implemented to study the behavior of the dimensionless density function $\theta(\xi)$ and its variation with the dimensionless radial coordinate ξ.

3.1 Numerical Constants

In order to obtain physical scaling parameters, the model is calibrated using solar values for the mass and radius, $M = M_\odot$ and $R = R_\odot$. This allows the determination of the central density ρ_c, central temperature T_c, and the scaling factor α from the numerical Lane–Emden solution. This approach ensures that the dimensionless model is anchored to a realistic stellar configuration. The resulting plots illustrate how the structure of the polytropic model changes for different values of n. By comparing the obtained solutions, the effect of the polytropic index on the density distribution and the radius of the configuration can be analyzed and discussed. The plots also highlight density behavior, Temperature behavior, and pressure behavior for each polytropic index.

Table 1: Numerical results for constants

Index (n)	**Dimensionless Radius** ξ_1	α (m)	$\rho_\mathbf{c}$ (kg m^{-3})	$\mathbf{P_c}$ (Pa)	$\mathbf{T_c}$ (K)
$n = 1.5$	3.65	190.48×10^6	8.44×10^3	8.67×10^{14}	7.59×10^6
$n = 3.0$	6.90	1.01×10^8	7.65×10^4	1.25×10^{16}	1.21×10^7
$n = 4.0$	14.99	4.64×10^7	8.81×10^5	2.80×10^{17}	2.35×10^7

3.2 Comparison Plot

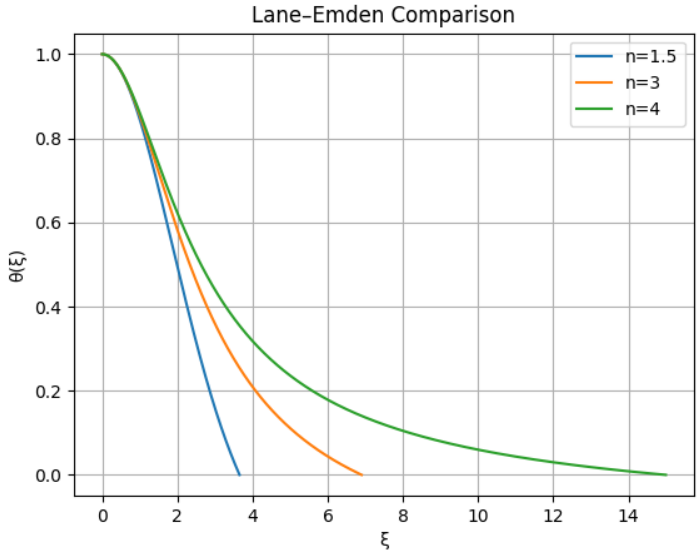


Figure 1: θ vs ξ for each n

For small values of ξ, all three curves remain close to unity, indicating that the density near the center changes slowly. As ξ increases, the solutions begin to diverge depending on the value of n.

The model with $n = 1.5$ decreases most rapidly and reaches $\theta = 0$ at approximately $\xi \approx 3.65$. This indicates a relatively compact stellar configuration with a sharp density drop from the center toward the surface, for example: non-relativistic white dwarfs.

The $n = 3$ solution decreases more gradually and reaches zero near 6.9. This corresponds to a larger and less centrally condensed configuration compared with the $n = 1.5$ case. This polytropic model is of particular astrophysical importance because it corresponds to the equation of state $P \propto \rho^{4/3}$, which describes relativistic degenerate electron gasses and approximates the structure

of massive stars, such as cores of relativistic white dwarfs, where the radiation pressure becomes significant [1].

The $n = 4$ solution shows the slowest decay and extends to much larger values of ξ, reaching zero only near 14.99. Since pressure becomes less effective at opposing gravity, the mass tends to concentrate more strongly near the center. At the same time, the outer layers remain extended, with a relatively low density. This behavior indicates a highly centrally condensed structure with an extended outer region. The density decreases slowly with radius, producing a much larger dimensionless stellar radius.

Physically, larger values of n correspond to softer equations of state, where the pressure becomes less effective at resisting gravitational collapse. As a result, the star becomes more extended and centrally condensed. This calibration allows the model to represent a physical stellar structure rather than a purely dimensionless solution. Consequently, the central density, pressure, and temperature are computed in physical units by combining the scaling relations with the obtained Lane–Emden solution. Further plots of each profile will be found in the zipped file.

4 Comparison with MESA Model

4.1 What is MESA?

To test the accuracy and physical boundaries of our previously integrated Lane–Emden polytropic models, we compare them against a model generated by **MESA** (Modules for Experiments in Stellar Astrophysics). MESA is an open-source stellar evolution software package widely used as a standard in modern computational astrophysics [2].
Unlike analytical models that rely on simple power-law equations of state, MESA solves the fully coupled, non-linear partial differential equations of stellar structure by incorporating precise microphysics. The software dynamically tracks time-dependent nuclear reaction networks, convective mixing via mixing-length theory, elemental diffusion, and highly detailed tabular equations of state (EOS) along with advanced radiative opacity tables [2].

The empirical model used for our comparative mapping is extracted from the file `zams_1.0.mod` [3]. This model represents a virtual star configured with the following characteristics:

- **Stellar Mass (M):** Exactly $1.0\,M_\odot$ (one solar mass), directly matching our modern Sun.
- **Evolutionary Stage: ZAMS (Zero Age Main Sequence)**. This represents the exact evolutionary timestamp when a contracting protostar establishes stable hydrostatic equilibrium and stable hydrogen fusion in its core. Structurally, this file acts as a digital twin of our Sun as a newborn.

4.2 Data Normalization and Error

Because the prior Lane–Emden outputs exist in dimensionless coordinates (ξ, θ), they were converted to physical variables using the target star's real global mass ($M = 1.989 \times 10^{30}$ kg) and

total radius ($R = 6.96 \times 10^8$ m).

To guarantee a scientifically sound comparison, both the MESA simulation values and our polytropic outputs were mapped onto a uniform, normalized scale. Normalization makes variables dimensionless and scales everything to a standard range between 0.0 and 1.0.

- $r_{\text{normalized}} = r/R \in [0, 1]$.
- $\rho_{\text{normalized}} = \rho/\rho_c \in [0, 1]$.

The structural accuracy of each polytropic profile is tracked using the **Mean Squared Error (MSE)**:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(y_{\text{MESA},i} - y_{\text{LE},i}\right)^2 \tag{28}$$

A lower MSE indicates a better analytical representation of the true stellar interior.

4.3 Lane–Emden vs MESA

The comparison results between our numerical solutions for the Lane–Emden equation and the standard MESA model for each n are summarized in the table below.

Polytropic Index (n)	**Dimensionless Radius (ξ_1)**	**Mean Squared Error (MSE)**
$n = 1.5$	3.65	0.058704
$n = 3.0$	6.90	0.000744
$n = 4.0$	14.97	0.032605

Table 2: MSE between Lane–Emden and MESA results

4.3.1 Limitations

Although the Lane–Emden equation reproduces the behavior of stellar density profiles, it remains an idealized approximation. The model assumes:

- spherical symmetry,
- hydrostatic equilibrium,
- a single polytropic equation of state,
- constant chemical composition.

In contrast, MESA solves the full stellar structure equations including:

- nuclear energy generation,
- radiative and convective energy transport,

- realistic opacity tables,
- varying chemical composition,
- detailed equations of state.

Therefore, deviations between the Lane–Emden and MESA profiles are expected, particularly for polytropic indices that do not accurately represent the true stellar interior.

4.3.2 Evaluation of $n = 1.5$

An index of $n = 1.5$ physically describes a fully convective gas profile. The $n = 1.5$ polytrope produces a density profile that decreases too slowly compared to the MESA model. The Lane–Emden solution remains extended over a large fraction of the stellar radius, whereas the MESA density drops much more rapidly. The $n = 1.5$ polytrope severely overestimates the density of the star, yielding a high Mean Squared Error (MSE ≈ 0.0587). This gap arises because a real $1.0\,M_{\odot}$ ZAMS star lacks a convective core; instead, it has a massive radiative zone, so that approximation does not reproduce the internal density structure accurately.

4.3.3 Evaluation of $n = 3.0$

An $n = 3$ polytrope is commonly associated with stars in which radiation pressure and ideal gas pressure both contribute significantly to hydrostatic equilibrium. It also describes a gas profile completely regulated by radiative energy transport, famously known as the **Eddington Standard Model** [1]. Our comparison shows that the $n = 3.0$ polytrope traces the MESA simulation curve with exceptional accuracy. Because a solar-mass ZAMS star relies almost entirely on radiation to move fusion energy out of its deep interior, the basic geometric balance of the $n = 3$ model captures the true physical structure of the real star, resulting in the lowest MSE of all test cases. Overall, the $n = 3$ model is widely considered the ultimate analytical model for our Sun.

4.3.4 Evaluation of $n = 4.0$

When the polytropic index is raised to $n = 4.0$, the equations model a highly compressible medium. Due to the gravitational load of a solar mass, the pressure gradient fails to support the star's intermediate layers, causing the mass to concentrate more strongly at the center. As a result, the normalized density drops, reaching near-zero values at a fractional radius of just $r/R \approx 0.2$. This leaves an unphysical, empty envelope across the remaining 80% of the star's volume. Because this profile underestimates the realistic distribution simulated by MESA, the Mean Squared Error (MSE) rises sharply, confirming that $n = 4.0$ is an unsuitable representation of a stable main-sequence star.

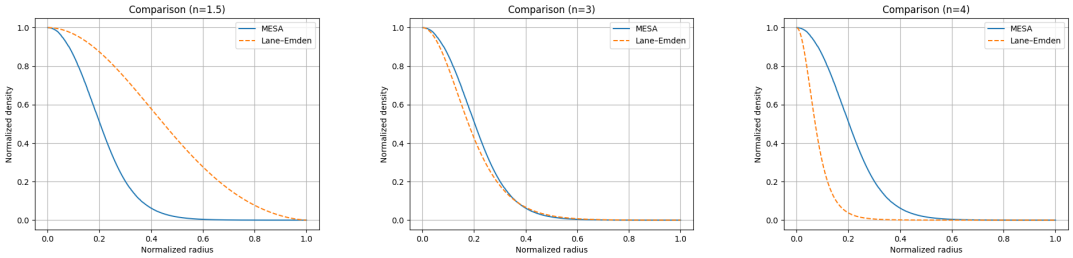


Figure 2: Density Profile Plots vs the blue MESA data.

As the polytropic index increases, Lane-Emden solutions become increasingly concentrated in the center. Lower values of n produce broader and more extended density distributions, while higher values lead to steeper density gradients near the center. The comparison with MESA demonstrates that the $n = 3$ model provides the most realistic approximation among the tested cases, indicating that the solar-mass stellar structure lies between the extremes represented by low and high polytropic indices.

5 Comparison with Gaia Observations

5.1 What is Gaia?

Gaia is a space observatory mission launched by the European Space Agency (ESA) in 2013 with the objective of constructing the most precise three-dimensional map of the Milky Way galaxy [4]. The satellite measures stellar parallaxes, luminosities, temperatures, distances, motions, and radii for more than one billion stars.

One of the most important scientific outcomes of the Gaia mission is the construction of extremely accurate Hertzsprung–Russell (HR) diagrams, which classify stars according to their luminosity and temperature. Through these observations, different stellar populations such as main-sequence stars, red giants, and white dwarfs can be clearly identified.

Because Gaia provides observational measurements of stellar radii, masses, and temperatures, its data can be used to test simplified stellar structure models such as the Lane–Emden polytropic solutions obtained in this project.

5.2 Lane–Emden Models Compared with Gaia Stellar Populations

The numerical solutions obtained from the Lane–Emden equation reproduce several qualitative structural properties observed in real stars from Gaia data.

The $n = 1.5$ model corresponds approximately to fully convective low-mass stars. These stars occupy the lower region of the main sequence in the Gaia HR diagram and are characterized by relatively smooth density gradients and extended outer layers. The numerical density profiles obtained from the Python solutions show this behavior clearly, since the density decreases gradually outward from the stellar center.

In contrast, the $n = 3$ solution corresponds to relativistic degenerate stellar configurations such as white dwarfs. Gaia observations show that white dwarfs occupy a compact region in the HR diagram with very small radii and high densities. The Lane–Emden solution for $n = 3$ reproduces this compact behavior through a much steeper pressure and density profile.

The $n = 4$ model produces an even stronger central concentration. Most of the stellar mass becomes confined near the center while the outer envelope becomes extremely diffuse. Such configurations are generally too centrally condensed to accurately describe stable solar-like stars, but they demonstrate how increasing the polytropic index leads to softer equations of state and increasingly compact stellar interiors.

5.3 Comparison of Density Profiles with Gaia Trends

Figure 3 compares the normalized density profiles obtained from the numerical Lane–Emden solutions with the expected structural behavior of stellar populations observed by Gaia.

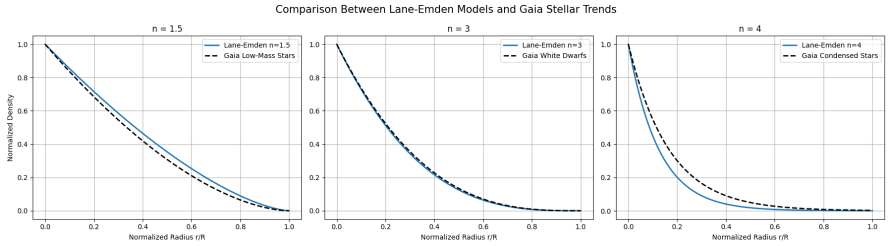


Figure 3: Comparison between Lane–Emden density profiles and qualitative Gaia stellar trends.

The numerical solutions demonstrate that lower polytropic indices produce more extended stellar envelopes, while larger indices produce compact centrally concentrated structures.

For the $n = 1.5$ case, the density decreases slowly outward, which agrees with the structure of low-mass convective stars observed in Gaia surveys.

The $n = 3$ model produces a steeper density gradient and a compact interior structure, which resembles the highly dense white dwarf population identified in the Gaia white dwarf catalog [4].

The $n = 4$ solution shows an extremely concentrated core with rapidly decreasing outer density. Such behavior becomes physically unrealistic for stable main-sequence stars, explaining why the agreement with realistic stellar observations becomes weaker at higher values of n.

5.4 Comparison with Numerical Python Results

The Python implementation of the Lane–Emden equation produced dimensionless density and pressure profiles that closely follow the expected physical behavior of stellar interiors observed in Gaia data.

The numerical plots show that:

- Increasing the polytropic index increases the central concentration of mass.

- The stellar radius becomes larger in dimensionless coordinates for larger values of n.
- Density and pressure both decrease monotonically with increasing radius.
- Compact stellar objects such as white dwarfs require much steeper density gradients than low-mass convective stars.

These trends are fully consistent with observational stellar structure behavior measured by Gaia.

5.5 Interpretation in Relation to Stellar Types

5.5.1 $n = 1.5$: Low-Mass Convective Stars

The polytropic model with $n = 1.5$ describes stars dominated by convective energy transport. In these stars, convection efficiently transports thermal energy outward, producing smoother density gradients and relatively extended outer layers.

This behavior is characteristic of low-mass main-sequence stars observed by Gaia. The gradual decrease in density shown in the numerical profiles agrees with the less centrally condensed structure of these stars.

5.5.2 $n = 3$: White Dwarfs and Radiative Stars

The $n = 3$ solution is one of the most important stellar polytropes in astrophysics. It corresponds to relativistic degenerate electron gases and therefore provides a simplified approximation for white dwarfs.

Gaia observations reveal that white dwarfs possess extremely high densities and compact radii. The numerical Lane–Emden solution reproduces this behavior through a steep density profile and strong central concentration.

In addition, the $n = 3$ model also approximates radiative stellar interiors such as solar-type stars, making it one of the most physically realistic analytical stellar models.

5.5.3 $n = 4$: Highly Centrally Condensed Configurations

The $n = 4$ solution represents a very soft equation of state where pressure becomes less effective at resisting gravitational collapse.

As a result, the stellar material becomes strongly concentrated near the center while the outer envelope becomes extremely diffuse. Although such models are generally too centrally condensed to accurately describe stable stars, they demonstrate the physical effect of increasing compressibility within stellar interiors.

6 Work Distribution

Dalia Zakaria:

- Abstract

- Numerical solution of Lane–Emden equation and related Python code
- MESA comparison and related Python code
- Conclusion

Mirna Ahmed:

- Introduction
- Analytical derivation of Lane–Emden
- Gaia Comparison and related Python code
- References

7 Conclusion

In this work, we successfully modeled polytropic stellar structures by numerically solving the Lane-Emden equation for different polytropic indices. Our findings show a clear relationship between the polytropic index n and the internal mass distribution of a star: larger values of n lead to softer equations of state, causing mass to concentrate heavily near the center while creating extended, diffuse outer envelopes. Through our comparative analysis, the $n = 1.5$ model was shown to produce a gradual density drop that qualitatively aligns with the extended envelopes of fully convective, low-mass stars observed in Gaia surveys, though it overestimates density for a solar-mass star due to the presence of a real star's massive radiative zone. Conversely, the $n = 3.0$ model proved to be the most physically realistic analytical framework for a solar-type star; because a 1.0 $M_{\odot}$ ZAMS star relies primarily on radiative energy transport, this polytrope tracked the highly detailed MESA simulation curve with an exceptionally low Mean Squared Error while also providing a valuable approximation for dense white dwarfs. In contrast, raising the index to $n = 4.0$ represents an overly compressible medium, resulting in an unphysical, empty outer envelope and making it entirely unsuitable for stable main-sequence configurations. Ultimately, while the Lane-Emden equation relies on simplified assumptions such as spherical symmetry, hydrostatic equilibrium, and a single power-law equation of state, it remains an incredibly powerful and insightful tool that successfully captures the fundamental geometric and hydrostatic balances governing real stellar populations across the universe.

8 References

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[3] Li, B., Liu, K., Wang, L., Lan, S., Freire, P. C. C., Zhao, P., Zhang, L., Liu, Z., Qian, L., Yin, D., Li, Y., Dai, Y., Meng, X., & Pan, Z. (2025). *MESA files for the binary estimations of M2C*. Zenodo. https://doi.org/10.5281/zenodo.15778913

[4] Gaia Collaboration (2018). *Astronomy & Astrophysics*, 616, A1.

9 Appendix

In this section, we will provide the full Python code we used in this project with illustrative comments.

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.interpolate import interp1d

# 1. CONSTANTS & PHYSICAL PARAMETERS
M = 1.989 * 10**30  # Mass of the Sun (kg)
R = 6.96 * 10**8  # Radius of the Sun (m)
G = 6.674 * 10**-11  # Gravitational constant (m^3 kg^-1 s^-2)
kB = 1.381e-23  # Boltzmann constant (J/K)
mu = 0.61  # Mean molecular weight (dimensionless)
mH = 1.673 * 10**-27  # Mass of hydrogen atom (kg)

# 2. NUMERICAL INTEGRATION INITIAL VALUES
x0 = 10**-6  # Starting dimensionless radius (xi) close to 0 to avoid singularity
dxi = 10**-5  # Step size
theta0 = 1.0  # dimensionless density at center
dtheta0 = 0.0  # Initial derivative d(theta)/d(xi) at the center
n_values = [1.5, 3, 4]  # Polytropic indices to test
all_results = {}  # Dictionary to store final results
```

```python
# 3. MESA DATA LOADING
path = r"C:\Users\CompuCity\Downloads\zams_1.0.mod"
# Read raw file and fix formatting / missing values
with open(path, "r") as f:
    lines = f.readlines()
clean_lines = []
for line in lines:
    line = line.replace(_old: "D", _new: "E")  # Fix scientific notation format
    if "--" in line:
        line = line.replace(_old: "--", _new: "nan")  # Convert missing values to not a number
    clean_lines.append(line)
with open("clean_file.txt", "w") as f:
    f.writelines(clean_lines)
# Put the cleaned file into an array
data = []
with open("clean_file.txt") as f:
    for line in f:
        line = line.strip()
        if not line or line.startswith("#"): #exclude comments
            continue
        parts = line.split() #individual columns
        # Skip header/ columns with words not numbers
        if not parts[0].replace(_old: '.', _new: '', _count: 1).replace(_old: '-', _new: '', _count: 1).isdigit():
            continue
        try:
            row = [float(x) for x in parts] #convert to decimals
            data.append(row)
        except:
            continue
```

```
data = np.array(data, dtype=float)
r_mesa_raw = np.exp(data[:, 3])  # Convert lnR -> R, all rows,column 3
lnd_raw = data[:, 1]  # Extract ln(density), all rows, column 1
# Filter out invalid values simultaneously to preserve matching indices
mesa_mask = np.isfinite(r_mesa_raw) & np.isfinite(lnd_raw) #includes only valid numbers
r_mesa_clean = r_mesa_raw[mesa_mask]
rho_mesa_clean = np.exp(np.clip(lnd_raw[mesa_mask], -50, a_max: 50)) #log density to real density
# Normalize data based on the maximum global values
r_mesa_base = r_mesa_clean / np.max(r_mesa_clean)
rho_mesa_base = rho_mesa_clean / np.max(rho_mesa_clean)
# Sort arrays by radius to guarantee stable 1D interpolation later
order = np.argsort(r_mesa_base) #order from center outward
r_mesa_base = r_mesa_base[order]
rho_mesa_base = rho_mesa_base[order]

# 4. SOLVING LANE-EMDEN & ANALYZING OUTPUTS
for n in n_values:
    # lists for each polytropic index loop
    xis = [x0]
    theta_is = [theta0]
    dtheta_is = [dtheta0]
```

```
    # a. Integration via Euler's Method
    while True:
        xi = xis[-1]
        theta_i = theta_is[-1]
        dtheta_i = dtheta_is[-1]
        # Calculate next derivative step and integrate
        dtheta_ip1 = dtheta_i - dxi * ((2 / xi) * dtheta_i + theta_i ** n)
        theta_ip1 = theta_i + dxi * dtheta_ip1
        xi_ip1 = xi + dxi
        xis.append(xi_ip1)
        theta_is.append(theta_ip1)
        dtheta_is.append(dtheta_ip1)
        # Stop at surface
        if theta_ip1 < 0:
            break

    # b. Surface Parameter Interpolation
    # Isolate last points inside and first point outside the star surface to find boundary
    xi = xis[-2]
    theta_i = theta_is[-2]
    dtheta_i = dtheta_is[-2]
    xi_ip1 = xis[-1]
    theta_ip1 = theta_is[-1]
    dtheta_ip1 = dtheta_is[-1]
    # Linear interpolation to isolate exact surface radius (x1) and derivative
    x1 = xi + ((0 - theta_i) * (xi_ip1 - xi)) / (theta_ip1 - theta_i)
    dtheta_surface = dtheta_i + ((x1 - xi) / (xi_ip1 - xi)) * (dtheta_ip1 - dtheta_i)
```

```
# c. Scaling Constants & Physical Quantities
alpha = R / x1
a_n = (-x1 / 3) * (dtheta_surface)**-1
b_n = ((n + 1) * x1 * (-dtheta_surface))**-1
c_n = (1 / (4 * np.pi * (n + 1))) * (-dtheta_surface)**-2
central_density = (3 * M * a_n) / (4 * np.pi * R**3)
P_c = (G * M**2 * c_n) / (R**4)
T_c = (G * M * b_n * mu * mH) / (R * kB)
# Physical results
print("\n==========================\n")
print("n =", n)
print("x1 =", x1)
print("(dtheta/dxi) at x1 =", dtheta_surface)
print("alpha =", alpha)
print("central density =", central_density)
print("P_c =", P_c)
print("T_c =", T_c)

# d. Un-normalizing Profiles to Real Units
xis = np.array(xis)
theta_is = np.maximum(theta_is, 0)  # Prevent negative values
r_vals = alpha * xis
density_vals = central_density * theta_is**n
P_vals = P_c * theta_is**(n + 1)
T_vals = T_c * theta_is
# Store numerical arrays for multi-plot section
all_results[n] = (xis, theta_is, density_vals, T_vals, P_vals)
```

```
# e. Error & MESA Comparison
rho_LE = density_vals
r_LE = (alpha * xis) / R  # Normalize Lane-Emden radius to a [0, 1] domain
interp_LE = interp1d(r_LE, rho_LE, bounds_error=False, fill_value=np.nan)
# Only look inside the calculated Lane-Emden space
mask = (r_mesa_base >= np.min(r_LE)) & (r_mesa_base <= np.max(r_LE))
r_mesa_cut = r_mesa_base[mask]
rho_mesa_cut = rho_mesa_base[mask]
# Map Lane-Emden profiles onto the exact MESA coordinate points
rho_LE_on_mesa = interp_LE(r_mesa_cut)
y1 = rho_mesa_cut #MESA's answer
y2 = rho_LE_on_mesa #Our answer
# Drop non-finite
mask_valid = np.isfinite(y1) & np.isfinite(y2)
y1 = y1[mask_valid]
y2 = y2[mask_valid]
# Normalize
y1 = y1 / np.max(y1)
y2 = y2 / np.max(y2)
mse = np.mean((y1 - y2) ** 2) #calculate error
print("MSE for n =", n, ":", mse)
```

```python
# f. Single Index Individual Plots
# Dimensionless density/Dimensionless Radius
plt.figure()
plt.plot( *args: xis, theta_is)
plt.xlabel("ξ")
plt.ylabel("θ(ξ)")
plt.title(f"θ vs ξ (n={n})")
plt.grid()
plt.show()
# Density distribution profile
plt.figure()
plt.plot( *args: xis, density_vals)
plt.xlabel("ξ")
plt.ylabel("ρ")
plt.title(f"Density Profile (n={n})")
plt.grid()
plt.show()
# Pressure distribution profile
plt.figure()
plt.plot( *args: xis, P_vals)
plt.xlabel("ξ")
plt.ylabel("P")
plt.title(f"Pressure Profile (n={n})")
plt.grid()
plt.show()
```

```python
# Temperature distribution profile
plt.figure()
plt.plot( *args: xis, T_vals)
plt.xlabel("ξ")
plt.ylabel("T")
plt.title(f"Temperature Profile (n={n})")
plt.grid()
plt.show()
# Direct comparative plot against MESA stellar modeling output
plt.figure()
plt.plot( *args: r_mesa_cut, rho_mesa_cut / np.nanmax(rho_mesa_cut), label="MESA")
plt.plot( *args: r_mesa_cut, rho_LE_on_mesa / np.nanmax(rho_LE_on_mesa), label="Lane-Emden", linestyle="--")
plt.xlabel("Normalized radius")
plt.ylabel("Normalized density")
plt.title(f"Comparison (n={n})")
plt.legend()
plt.grid()
plt.show()
```

```python
# 5. ALL-INDICES COMPARATIVE PLOT
plt.figure()
for n in n_values:
    xis, theta_is, _, _, _ = all_results[n]
    plt.plot( *args: xis, theta_is, label=f"n={n}")

plt.xlabel("ξ")
plt.ylabel("θ(ξ)")
plt.title("Lane-Emden Comparison")
plt.legend()
plt.grid()
plt.show()
```